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A Dynamic Attribute Satiation Model of Variety-Seeking Behavior

LEIGH McALISTER*

This paper presents a model of individual consumer choice for separate choice occasions. Contrary to the notion that each choice is essentially independent of its predecessors, dependence is proposed as the key to variety-seeking behavior. As an individual's consumption history evolves, the pattern of attribute accumulations changes, leading to preference for items rich in different attributes at different points in time. The model is operationalized using soft drink consumption diaries, and predicts choices better than a model that ignores choice dependence.

An excerpt from a soft drink consumption diary indicates that the subject drank a Coke on Monday, a Coke on Tuesday, a Dr. Pepper on Wednesday, a 7-Up on Thursday, a Pepsi on Friday, and a club soda on Saturday. Further questioning revealed that Dr. Pepper was selected on Wednesday because it was offered that day at an especially low price. Seven-Up was chosen Thursday as a change of pace. The subject wanted a Coke on Friday but settled for a Pepsi because Coke was not available. On Saturday night, the subject chose club soda to mix in her Scotch drink.

This pattern of switching among brands is not atypical for frequently purchased, nondurable goods. Switching can be induced by the manipulation of marketing variables, by the accessibility of the product, by the situation in which the product is consumed, or by the desire for variety. It is switching for the sake of variety with which this paper is concerned. In particular, it is proposed that such switching is not completely random. A deterministic process that would lead to variety seeking is proposed, modeled, and empirically tested. The process differs from traditional models in its consideration of the impact of previous selections.

The hypothesized effect of consumption history on variety seeking is based on three assumptions. Following Lancaster (1971), it is assumed that items can be represented by the values they take on for their constituent attributes. Similarly, a collection of items can be represented by the sum of values, across items in the collection, on those constituent attributes. A consumption history, which is a col-

lection of items, can therefore be represented by the attribute accumulations or attribute "inventories" it generates.

Consider the simplified example in Tables 1 and 2. Individual soft drinks are described by the two attributes, fruit flavor and caffeine. At any point in time, the consumption history can be summarized by the amounts of these two attributes that have been accumulated. On Day 1, only one Coke was consumed, so the accumulated inventories would be 2 units of fruit flavor and 9 units of caffeine. The consumption of another Coke on Day 2 would raise the inventories by 2 and 9 units, respectively. After the consumption of a Dr. Pepper on Day 3, the fruit flavor inventory would be increased by 5 units and the caffeine inventory by 7 units, and so on.

The second assumption concerns the continual depletion of the inventories through physiological processing or forgetting. This depletion can be conceptualized as the discounting of older consumption experiences. That is, the soft drink that a subject consumed a week ago has a smaller residual impact on her current attribute inventories than one she consumed yesterday. Similarly, that soft drink she consumed yesterday would have a smaller residual impact than one she consumed one hour ago. Figure A describes the ups and downs of the caffeine inventory generated by the consumption history in Table 2. From day to day, the inventory drops by half to represent the physiological processing or forgetting of the attribute.¹ With each consumption, there is a discrete jump upward in the level of the inventory equal in size to the amount of the attribute in the soft drink consumed.

Finally, it is assumed that there is a decreasing marginal relationship between the attribute inventories that would

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¹The fraction of the inventory that carries through from one day to the next might be any value from 0 to 1. In operationalizing the model, it was necessary to select a particular value. One half was chosen arbitrarily. This issue is discussed further in the section on "Operationalizing the DAS Model."

TABLE 1
SIMPLIFIED DESCRIPTION OF SOFT DRINKS

Soft drink	Fruit flavor	Caffeine
Coke	2	9
Dr. Pepper	5	7
7-Up	9	0
Pepsi	3	8
Club soda	0	0

result from the consumption of a particular item and preference for that item. Figure B depicts two possible decreasing marginal relationships:²

- A finite ideal point that lies within the achievable range of the attribute inventory (given by the dashed curve)
- A finite ideal point that lies beyond the achievable range of the attribute inventory (given by the broken curve)

Consider a single attribute. When the inventory of that attribute is very low, this assumption implies that preference for a unit of that attribute is at its highest. As the inventory of that attribute grows, preference per unit drops. Should the inventory reach its ideal level or point of satiation, the marginal impact of adding a unit of the attribute is zero. Beyond the point of satiation, preference decreases. This phenomenon explains why a cola might be very appealing at times (when the inventory of caffeine is low and thus the marginal impact on preference of adding to that inventory is relatively high) and less so at other times (when the inventory is high, implying that additions will have a relatively small marginal impact on preference).

The amalgamation of these three assumptions is a dynamic attribute satiation process, which holds that preference for an item at a point in time is a function of the preference contributions of that item's constituent attributes.

²A constant marginal relationship is also depicted in Figure B (the solid line). This relationship implies that each incremental unit of an attribute contributes equally to preference. It is reasonable to believe that some attributes will display such a relationship. They would not, however, contribute to the kinds of preference shifts of interest here.

butes. The preference contribution of each attribute is, however, a function of the consumption history (summarized by attribute inventories) and the point of satiation for that attribute. Since the configuration of attribute inventories can change dramatically as the consumption history evolves, it is not surprising that shifts in preferences among choice alternatives (and resulting variety seeking) are observed.

In this paper, the dynamic attribute satiation process is operationalized as an estimatable model. Soft-drink consumption diaries are used to infer individual specific parameterizations of that model. The ability of the proposed model to predict actual choices is shown to be superior to that of a model based on the assumption that each choice is independent of prior selections. The managerial implications of this finding are discussed.

LITERATURE REVIEW

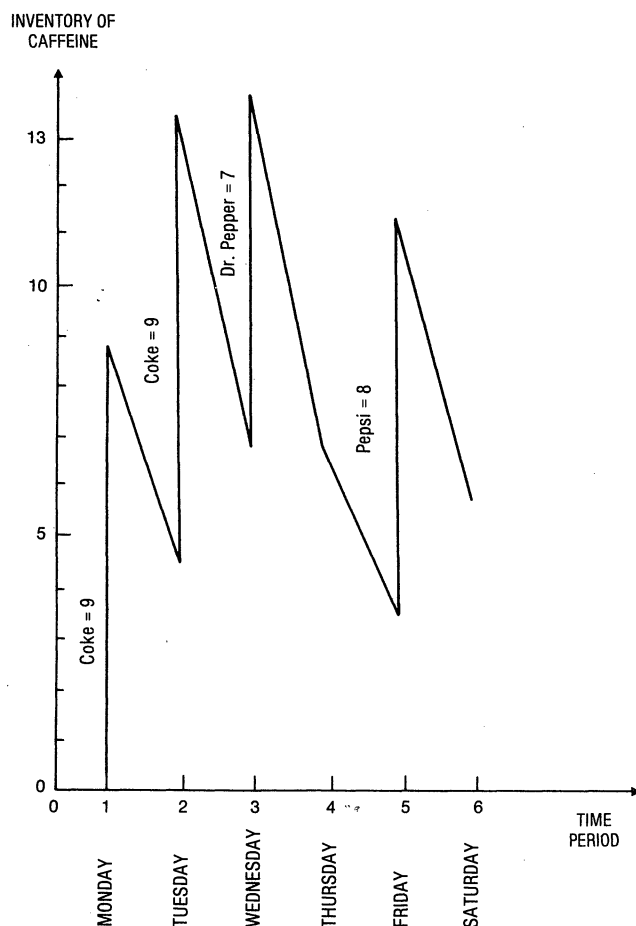
As mentioned earlier, brand switching is sometimes induced by the manipulation of marketing variables (price, product design, promotion, distribution) and sometimes by changes in situational variables. Early studies of brand loyalty (McConnell 1968; Tucker 1964) and a similar study in social psychology (Brickman and D'Amato 1975) controlled for these factors and still reported switching. In those studies, subjects were asked to make repeated choices from a set of unfamiliar items. Two distinct phases of switching behavior were apparent in the data. Initially, subjects systematically explored the items. Later in the experiment, subjects tended to alternate among the elements of their favored subset of the items. Much of the consumer behavior literature on variety seeking, novelty seeking, innovativeness, and so on focuses on switching like that of the first phase (e.g., Hirschman 1980; Robertson 1971; Venkatesan 1973). This phase can be distinguished³ from the later one by the information acquisition motive. It is switching like that of the second phase (switching among familiar items

³Keon (1980) would disagree. He argues that switching among familiar items results from the desire to refresh one's memory and assure oneself that the usual selection is superior.

TABLE 2
HYPOTHETICAL CONSUMPTION HISTORY AND RESULTING INVENTORY OF CAFFEINE

Time period (t)	Soft drink consumed in period t	Contents of soft drink consumed in period t		Caffeine inventory		
		Fruit flavor	Caffeine	Inventory from period t-1	Discounted inventory from period t-1	Inventory at the end of period t
1	Coke	2	9	—	—	9.0
2	Coke	2	9	9.0	4.5	13.5
3	Dr. Pepper	5	7	13.5	6.8	13.8
4	7-Up	9	0	13.8	6.9	6.9
5	Pepsi	3	8	6.9	3.5	11.5
6	Club soda	0	0	11.5	5.8	5.8

FIGURE A
INVENTORY OF CAFFEINE



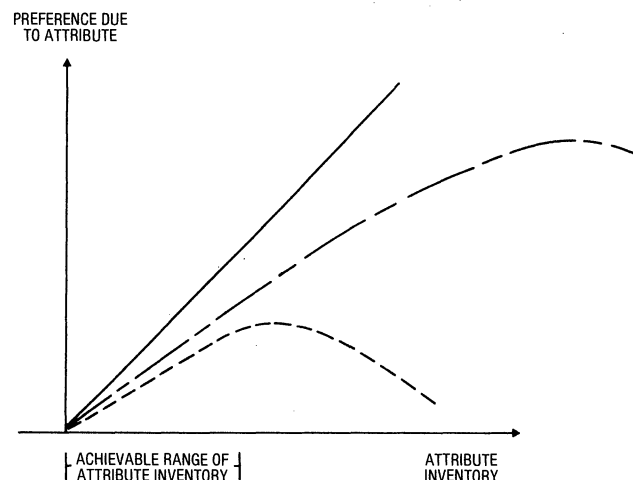
that is done simply for “variety”) with which this paper deals.

Bass, Pessemier, and Lehmann (1972) studied choice from a familiar product class, controlling the impact of marketing and situational variables. Subjects in their study chose the item they reported as their favorite only about half the time. Bass et al. attributed the switching to and among reportedly less preferred items to “a stochastic component of choice which arises because of variety seeking” (1972, p. 538). This line of reasoning gave rise to Bass’ (1974) “theory of stochastic preference and brand switching.” According to that theory, relative preferences dictate the proportion of times each item is chosen. Any particular selection, however, is at the mercy of a “stochastic element in the brain.” Blin and Dodson (1980) propose a specific functional relationship between the frequency with which an item will be chosen and the importance weights in an individual’s linear compensatory preference function. In a variation on this theme, Huber and Reibstein (1978) propose that the item selected is a deterministic function of parameters of the preference function of the person doing the choosing. Those parameters, however, are assumed to change in a random fashion.

Under the theory of stochastic preference, variety seeking

FIGURE B

RELATIONSHIP BETWEEN ATTRIBUTE INVENTORY AND THE PREFERENCE CONTRIBUTION OF THAT ATTRIBUTE FOR FINITE AND INFINITE IDEAL POINTS



NOTE: Preference for an attribute with a finite ideal point that lies within the achievable range of the attribute inventory ————
 Preference for an attribute with a finite ideal point that lies beyond the achievable range of the attribute inventory ————
 Preference for an attribute with an infinite ideal point ————

is viewed as nonunderstandable. Other paradigms allow explanation of the phenomenon. Predominant among the explanations is the notion of “optimal arousal” (Berlyne 1960; Hansen 1972). Individuals are hypothesized to select that collection of items which will provide just the right amount of stimulation or arousal. Optimal arousal is frequently conceptualized as a single peaked relationship (Coombs and Avrunin 1977) between preference and the stimulation provided by the collection selected. McAlister (1979) modeled the stimulation provided by the collection as a function of the stimulation provided by attributes of items in the collection. Her model predicts the collection of items a person would select at a point in time.

Jeuland (1978) modeled variety-seeking behavior across time. He essentially proposed item- (i.e., brand-) specific stimulation optima. He further proposed “inventories” of item stimulation that evolve dynamically in a manner analogous to the evolution of attribute inventories. Implicit in this formulation is the assumption that the consumption of one item has no impact on the inventories of other items. As pointed out by Hagerty (1980), this ignores the effect of item similarity. Under Jeuland’s model, the preference for Coke should not be affected by either of the following consumption histories: (1) Pepsi, Dr. Pepper, RC Cola, Pepsi, or (2) 7-UP, 7-UP, 7-UP, Mountain Dew, Sprite. Changes in preference result from changes in the relevant inventory, and in either of these cases, the inventory of Coke would be zero.

In the proposed dynamic *attribute* satiation process, inventories of attributes rather than inventories of items are accumulated. The first consumption history given above would generate a large inventory of caffeine and a small inventory of fruit flavor. The second consumption history

would generate a large inventory of fruit flavor and a small one of caffeine. The evaluation of the caffeine and fruit flavor contributions of a Coke (and hence the evaluation of Coke itself) would probably be very different with the first consumption history than it would be with the second.

OPERATIONALIZING THE DAS MODEL

Consistent with models in the traditions of Fishbein (1963) and Lancaster (1971), the Dynamic Attribute Satiation (DAS) model relates preference for an item to the preference contributions of the attributes of that item. Assuming there are no interactions among the attributes,⁴ preference for an item will be defined as the sum of the contributions to preference made by each of the attributes, that is:

$$DAS_{Tk} = \sum_{j=1}^J P_{Tkj} \quad (1)$$

where

DAS_{Tk} = preference for item k at time T assigned by the Dynamic Attribute Satiation model

J = the number of attributes over which items are described

P_{Tkj} = contribution of attribute j to the level of satisfaction that would result from consuming item k at time T

The contribution of attribute j to satisfaction from consuming item k at time T (i.e., P_{Tkj}) is assumed to be a function of the inventory of attribute j at time T , the ideal level of attribute j , and the amount of attribute j in item k . It was argued that the particular functional relationship should be marginally decreasing with a single optimum at the ideal point. A quadratic relationship of the form presented in Equation 2 displays these properties and is easily estimated:⁵

$$P_{Tkj} = w_j [(I_{Tj} + X_{kj}) - \hat{X}_j]^2 (-1) \quad (2)$$

where

P_{Tkj} = contribution of attribute j to the level of satisfaction that would result from consuming item k at time T

w_j = importance weight for attribute j

I_{Tj} = inventory of attribute j at time T

X_{kj} = amount of attribute j in item k

⁴If there is reason to believe, a priori, that interactions exist between certain attributes, a new attribute, defined as the product of the interacting attributes, can also be used. With this augmentation of the attribute set, the interaction is accounted for and the proposed model is still valid.

⁵The estimation procedure to be used is LINMAP. This algorithm selects values for the importance weights and ideal points to most closely replicate the original preference ordering. An option available to the algorithm is to set ideal point for a given attribute at infinity. In that case, Equation 2 degenerates into a simple linear relationship. (See Srinivasan and Shocker 1973, pp. 345–346.)

$\hat{X}_j$ = ideal level for attribute j

Note that when the current inventory of attribute j plus the amount of attribute j in item k ($I_{Tj} + X_{kj}$) exactly equals the ideal for attribute j ($\hat{X}_j$), the term which is squared equals zero, forcing P_{Tkj} to also equal zero. The more $I_{Tj} + X_{kj}$ differs from $\hat{X}_j$, the more negative P_{Tkj} becomes—that is, preference peaks at the ideal point. To either add to or diminish the inventory reduces overall preference.

Attribute Inventory

The attribute inventory is hypothesized to dwindle continuously and experience discrete increments each time an item containing that attribute is consumed (Figure A). A similar process was proposed by Little and Lodish (1969) for the cumulative impact of advertising and by Jeuland (1978) for the evolution of “experience” with a brand. The particular functional form used here to convert a consumption history (X_{kj} , $t = 1, 2, \dots, T$) into an inventory is:

$$I_{Tj} = \sum_{t=1}^T \lambda_j^{T-t} X_{ktj} \quad (3)$$

where

I_{Tj} = inventory of attribute j at time T

λ_j = inventory retention factor for attribute j ,
 $0 \leq \lambda_j \leq 1$

k_t = item chosen at time t

X_{ktj} = amount of attribute j in the item chosen at time t

The speed with which the inventory dwindles is an inverse function of the inventory retention factor, λ_j . It seems reasonable that this factor might differ across individuals and across attributes. For instance, one might expect $\lambda_{\text{thirst quenching}}$ for athletically active persons to be smaller than $\lambda_{\text{thirst quenching}}$ for more sedate persons, indicating a more rapid dwindling of the inventory of “thirst quenchingness” for athletically active persons than for the more sedate. (In general, the λ 's for heavy users of a product class might be expected to be lower than λ 's for light users.) Similarly, one might expect $\lambda_{\text{calories}} > \lambda_{\text{carbonation}}$ for an individual indicating that the human body divests itself of carbonation more rapidly than calories.

Unfortunately, the estimation procedure employed requires the a priori specification of values for the λ_j 's. Receiving no guidance on the selection of such values from those who have proposed similar processes (Jeuland 1978; Little and Lodish 1969), the value for λ_j was set arbitrarily at $1/2$ for all attributes and all individuals.⁶ This arbitrary

⁶Because of the nature of the data collected and the estimation procedure used, over an hour of CPU time was required to estimate the models presented in this paper. It would be informative to reestimate the models assuming different values for λ . Comparison of the predictive ability of such reestimated models would shed light on the nature of the inventory retention factor. However, because of the exorbitant data processing requirements, and because the results obtained with $\lambda = 1/2$ were sufficient to address the central hypothesis, the models were not reestimated.

selection will impede the predictive ability of DAS. However, the objective of this paper is to demonstrate that a process which considers the interdependence of choices has greater predictive ability than one which views each choice as independent of all others. If the process incorporating the interdependence (DAS) is able to make better predictions in spite of a handicap in parameter selection, the point will be made. Future research into the pattern of variation of these parameters should provide insights useful to researchers and managers and enhance the predictive ability of such models.

Parameters of the Model

Ideal Points. It is important to note that the ideal points in the DAS model, $(\bar{X}_j; j = 1, 2, \dots, J)$, do not necessarily reflect the amount of attribute j that one would like to find in a single choice alternative. Rather, they indicate the optimal level for the entire inventory of that attribute. As the inventory level of an attribute decreases, the amount of that attribute which is desired in a choice alternative increases.

Importance Weights. The difference between an inventory retention factor and an importance weight should be noted. An importance weight reflects the degree of disutility associated with being a given number of units from one's ideal level. Comparisons among importance weights allow one to make statements concerning the relative impact of each attribute *assuming that all else is held constant*. But all else is not held constant, and inventory retention factors determine the pattern of that inconstancy. For example, being one unit away from one's ideal level of thirst quenchingness might be less unpleasant than being one unit away from the ideal level of sweetness (importance weight for sweetness > importance weight for thirst quenchingness). However, if this individual's $\lambda_{\text{thirst quenchingness}}$ and $\lambda_{\text{sweetness}}$ were such that the inventory of thirst quenchingness depleted much more rapidly than the inventory of sweetness, alternatives offering relatively more thirst quenching ability than sweetness would probably be chosen, in spite of the fact that sweetness has a higher importance weight than thirst quenching ability.

Confounding of Parameter Estimates when Inventory is Ignored. If people do, in fact, evaluate choice alternatives relative to the current state of their attribute inventories, one might expect bias to be introduced by estimation procedures which do not account for those inventories. The Appendix shows that the failure to account for inventories will not bias importance weights but will bias ideal points. The direction of ideal point bias will be downward from the true ideal by an amount equal to the inventory at that point in time. For that instant, the bias in the parameter exactly makes up for the failure to include inventory in the stimulus-attribute matrix. At later points in time, the inventory will be at different levels and the parameter's bias will *not* exactly compensate. The managerial implications of this bias are explored in the "Discussion" section.

DATA

Data for the study were collected from October through December 1978. Twice in the course of this 81-day study (at the beginning and half way through), subject perceptions of stimuli on relevant attributes were elicited. Twice a week during the course of this study, subjects were asked to rank-order the stimuli from most preferred to least preferred and to report all items from the stimulus product class that they had consumed since they last gave data. Data from the first 50 days of the study are used to estimate parameters of different models. These estimated models are then employed to predict actual choices made during the last 30 days of the study. Statistics based on those predictions allow comparison of models.

Subject Population

The subjects in this study were 36 graduate and undergraduate students enrolled in the School of Business Administration at the University of Washington during fall quarter, 1978. Twenty-two percent of the subjects were females. No economic incentive was offered for participation in the study. Motivation for accurate reporting of information was provided by having the professor in a bi-weekly class begin each period by distributing data collection forms and requesting that serious thought be given to the task. Responses to a questionnaire administered at the end of the study indicate that subjects were neither intrigued nor irritated by the tasks⁷ and tended to report their preferences accurately.⁸

Stimuli

This study, like Bass et al.'s (1972), deals with preferences for and choices among soft drinks. The 10 stimuli selected were: Coke, Diet Pepsi, Dr. Pepper, Mountain Dew, Pepsi, Royal Crown Cola, 7-Up, Sprite, Sugar Free 7-Up, and Tab. These soft drinks were reported in *Standard and Poor's Industry Survey* (1978) as having the 10 highest market shares, ranging from a share of 26.6 percent for Coke to 1.2 percent for Sugar Free 7-Up. The total of market shares for stimulus soft drinks is 69.2 percent, with no excluded soft drink receiving more than 0.9 percent of the market. These soft drinks include 6 colas and 4 noncolas and 3 diet drinks and 7 nondiet drinks.

The subjects were, on average, reasonably familiar with this product class. Their self-reported familiarity with the stimulus soft drinks, averaged across subjects and across soft drinks, is 2.8 on a scale on which 1 indicates that the

⁷Subjects self-reported their level of interest in the tasks on a 1 to 5 scale on which 1 indicated that they were quite interested in reporting preferences and perceptions and 5 indicated that they were irritated by requests to report perceptions and preferences. The average value of this variable across subjects was 3.03.

⁸Subjects self-reported the accuracy of their preference reports on a 1 to 5 scale on which 1 indicated that their preference reports were unrelated to their true preferences and 5 indicated that their preference reports exactly reflected their true preferences. The average value for this variable across subjects was 3.81.

subject had never tasted the soft drink and 5 indicates that the subject often drinks it.

The attributes selected to describe the stimulus soft drinks are the same as those used in Bass et al.'s (1972) study: carbonation, calories, sweetness, thirst quenching ability, and popularity with others.⁹ Seventy-two percent of the subjects indicated that the selected attributes were not the only factors they considered in selecting a soft drink. Other attributes that they mentioned as important include (in order of frequency of mention, most frequent first): taste, price, availability, caffeine, saccharine, can size, and nutrition.

Data Collection Instrument

The data collection instrument was made up of three documents. The first document, administered at the beginning of the study and half way through, elicited subject perceptions of attribute values for stimulus soft drinks on 6-point scales, where 0 indicated that the brand possessed virtually none of the attribute and 5 indicated that the brand possessed a great deal of the attribute. The order in which stimulus soft drinks were presented was randomized across subjects.

The second document, administered biweekly for the 11 weeks of the study, elicited a history of soft drink consumption since the subject had last given data. It also requested that the 10 stimulus soft drinks be rank-ordered from the brand a subject would most like to consume at that point in time to the brand she would least like to consume at that point in time. The order of presentation of soft drinks for the rank-ordering task was randomized across data collection occasions.

The final document was administered at the end of the study to survey subjects' attitudes about the study. It included questions regarding familiarity with the stimuli, involvement with the experimental tasks, reasons for selecting a soft drink other than the one which had most recently been reported as most preferred, and factors other than experimental attributes that influenced the subject's selection of soft drinks.

Model Estimation

The algorithm LINMAP was used to estimate parameters for DAS. Because not all subjects reported preferences at every possible data collection opportunity (due to class absences), the number of preference reports per subject in Days 1 to 50 of the study (the time period used to estimate parameters) ranges from 5 to 11. Their respective stimulus-attribute matrices were therefore made up from 50 to 110 stimuli (10 from each report, corresponding to the perceptions of the 10 stimulus soft drinks augmented by the inventories). This does not, however, provide the information

content of a single ranking of 50 to 110 items. Since no comparisons were made across report periods, it is impossible to say whether the item ranked third on one day is more or less preferred than one ranked fifth on another day. The information content of the 50 to 110 stimuli was estimated,¹⁰ and the measures were considered sufficient to confidently estimate the 12 parameters (an ideal point and importance weight for each of the six attributes).

Two versions of an alternative model that treated each choice as independent of all others were also estimated using LINMAP. Under this paradigm, attribute inventories were not considered. Therefore the stimulus set facing subjects on each data collection occasion was considered identical. Their multiple reports of preference rankings were then viewed as replications of the same task. The two versions of this alternative model were distinguished by the dependent variables used in their estimation: one was based on the ranking from the first preference report given by a subject, and the other was based on a ranking consistent with the average, across-preference reports for that subject.

RESULTS

Two types of tests were utilized to assess the performance of DAS relative to a model that does not incorporate the impact of past choices: (1) goodness-of-fit tests, which deal with the models' relative abilities to replicate the preference reports (from the first half of the study) that were used in their estimation; and (2) an external validity test, which compares the models with respect to their ability to predict actual choices made during the second half of the study.

The statistics used for the goodness-of-fit comparisons are the Kendall's taus reported by LINMAP in estimating the models' parameters. These statistics measure the correlation between the actual preference reports and the preference ranking predicted by the estimated model. They are compared in three different ways in Table 3: Column 1

¹⁰Two criteria were considered for measuring the information content of a rank-ordering. The first is the number of inherent paired comparisons: a ranking of k objects implies $k(k-1)/2$ paired comparisons. This measure multiple counts many pieces of information that are apparent via transitivity. Consider a ranking of four objects: A, B, C, and D. This ranking implies $4(3)/2 = 6$ paired comparisons ($A > B$, $A > C$, $A > D$, $B > C$, $B > D$, $C > D$; where $>$ means "is preferred to"). However, given that one knows $A > B$ and $B > C$, it is redundant to say that $A > C$ (assuming transitivity). The second proposed measure disregards the redundant paired comparisons. It holds that the number of pieces of information in a ranking of k objects is $k-1$ ($1st > 2nd$, $2nd > 3rd$, . . . , $k-1st > kth$).

According to Green and Srinivasan (1978), in selecting the number of stimuli necessary for accurate parameter estimation, one should consider the ratio of (number of stimuli)/(number of parameters estimated). "Number of stimuli" in that calculation is analogous to the number of objects in a single rank-ordering. The "number of parameters estimated" for a given subject in this study is 12 (an ideal point and an importance weight for each of the 6 attributes). The relevant ratio using the first measure of information content runs from a lower bound of $21/12 = 1.75$ for the subject providing only 5 preference reports to an upper bound of $31/12 = 2.67$ for those subjects providing 11 preference reports. Using the second measure of information content, the ratio runs from $46/12 = 3.83$ to $100/12 = 8.33$.

⁹For purpose of analysis a sixth attribute, cola/noncola, was also used. Because the value of that dichotomous attribute was obvious for the stimulus soft drinks, it was not necessary to elicit subject perceptions concerning it.

TABLE 3
TESTS OF MODELS^a

Model	Goodness of fit			External validity
	Average τ (standard deviation)	Percent of τ 's > 0.8	Percent of τ 's > correlation of first preference report with market share	Percent of choices predicted correctly
Dynamic attribute satiation (DAS)	.75 (.19)	33	100	26
Independent choices, first preference	.67 (.16)	31	94	17
Independent choices, average preference	.64 (.15)	22	89	15

^an = 36

reports the average value for tau (τ) across subjects; Column 2 reports the percentage of the subjects for whom the value of τ is greater than 0.8; and Column 3 reports the percentage of subjects whose reported preferences correlate more highly with model predictions than with the average preference ranking (interpreting the study-based market shares as an average preference ranking). The measure of external validity reported in Column 4 is the percentage of actual choices made which were correctly predicted by the various models.

Among the three models estimated (DAS, Independent Choices based on the First Preference Report, and Independent Choices based on the Average Preference Report), DAS dominates on all measures. Tests for the statistical significance of the differences between the average τ for DAS and those of the two variants of the Independent Choices model are significant at the 0.01 level. The difference between the two variants of the Independent Choices model is not significant, even at the 0.1 level. These tests are summarized as follows:

	Difference	Average across 36 subjects	Standard deviation	τ -statistic
$\tau_{DAS} - \tau_{Indep. Choices 1st Pref.}$		.09	.19	2.80
$\tau_{DAS} - \tau_{Indep. choices Avg. Pref.}$		.11	.18	3.62
$\tau_{Indep. choices 1st Pref.} - \tau_{Indep. Choices Avg. Pref.}$		.02	.14	.85
$\tau_{35} (.01) = 2.44$				
$\tau_{35} (.1) = 1.31$				

DISCUSSION

From the reported statistics, one can infer that, for these subjects, DAS reproduces preferences and predicts choices better than a model that does not consider past choices. This can be taken as evidence that the inventorying process outlined in this paper is an important determinant of choice behavior. Under that assumption, several issues deserve further consideration.

It was shown that estimates of ideal points are biased when models are estimated without regard for attribute inventories. The interpretation of this bias depends on the wording of statements that elicit the rank-ordering used in estimating parameters. The two forms of questioning typically employed have to do with reporting "preferences at this point in time" or "preferences generally."

The first form of questioning will elicit biases equal to the current inventories of the attributes. Model reestimation at different points in time could very well yield radically different values for ideal points. (Perhaps it was observation of some such phenomenon that led Huber and Reibstein (1978) to hypothesize their random ideal points model.) Each estimated ideal point would be biased by an amount equal to the level of inventory at the time the rank-ordering was elicited. A model so estimated should do a good job of predicting choice behavior at that particular point in time. As inventories dwindle and are replenished, however, the preference ranking will very likely change.

The second form of questioning ("preferences generally") will elicit biases equal to average inventories of attributes. These biases—in contrast to those just discussed—should be relatively stable. Model reestimation at different points in time should yield similar values for ideal points. Unfortunately, there is no guarantee that this model will reflect the true preference ranking for any particular choice occasion. For example, on a given choice occasion one might have a very high inventory of sweetness, suggesting the choice of a nonsweet item. By the next choice occasion, the inventory of sweetness could have dwindled dramatically, making a very sweet item appealing. Preferences based on average inventory level would suggest the choice of a moderately sweet item: in fact, the one nonsweet item and one very sweet item were chosen.

The importance of attribute inventories suggests a new class of model parameters, namely, inventory retention factors, to be investigated. Although the values for all inventory retention factors for all subjects in this study were taken to equal 1/2, arguments have been offered concerning the likely variability of those parameters. In sum, it has been proposed that the inventory retention factors are determined by underlying physiological and/or psychological processes. For example, inventory of "sweetness" might be analogous to blood sugar level. The inventory retention factor would be, in that case, a function of the body's ability to metabolize sugar. Inventory retention factors for non-physiologically based attributes like "popularity with others" or "stylishness" might be a function of self-confidence, innovativeness, or similar psychological constructs.

Given these systemic roots for the factors, hypotheses concerning the patterns of values across populations and across time can be posited and tested.

The results of such studies should have far-reaching managerial implications. It will be important to eliminate or at least to understand the biases caused by ignoring inventories. Designing new products or repositioning old ones on the basis of ideal points derived from models that do not consider inventories could be dangerous for the reasons previously outlined. It could be similarly misleading to segment the population based on importance weights alone. While the weights are not biased by estimation procedures that ignore inventories (see Appendix), they alone do not tell the whole story. For example, sweetness might be twice as important to person A as to person B, but if their inventory retention factors indicated that person B's inventory would drop 6 times faster than that of person A, relative choice behavior might be counter to expectation based on importance weights alone.

Even more important than those specific implications, consideration of attribute inventories may provide the manager with a clearer understanding of the process by which consumers make choices. It would seem that this better understanding should aid the manager in all aspects of marketing her product.

DIRECTIONS FOR FUTURE RESEARCH

The Dynamic Attribute Satiation (DAS) Model is simply one more step in the process of explaining individual choice behavior. Consideration of the impact of situational variables in predicting individual choice has improved that prediction (e.g., Warshaw 1980). While this is surely a step forward, it is not the end of the process. In the Bass et al. (1972) study, situational factors were controlled and switching among brands was still observed. DAS was designed specifically to accommodate that nonsituationally based switching. The version of the DAS model empirically tested in this paper was able to expand prediction beyond that of a model which does not account for the impact of the consumption history. This is especially encouraging because the inventory retention factors, which are the key to encoding the effect of the consumption history in DAS, were only crudely approximated. Exploration of these parameters should further improve the predictive ability of the model and provide valuable insight into the process by which individuals make choices.

Further research is needed into the nature of the inventory retention factors and the sensitivity of predictive ability to a wide range of values for those parameters. Specific hypotheses linking inventory retention parameter values to demographic and psychographic variables should be explored. The estimation procedure used in this paper requires that values for the inventory retention factors be tested by trial and error. No convexity or even continuity properties can currently be attributed to the relationship between parameter values and predictive ability. Hence improvement can be noted, but global optima cannot be guaranteed. Per-

haps a reformulation of the inventory term (Equation 3) or the creative use of estimation procedures could solve this problem. Better estimation of parameter values should, in turn, lead to questions concerning the specific implications for different product classes and population segments.

The calculation of attribute inventories needs to be extended to acquisition of attributes from products overlapping the product class under study by one or more attributes. For example, the inventory of sweetness relevant to the selection of soft drinks is affected not only by sweetness from historical soft drink consumption, but also by historical consumption of cake, pie, fruit, and so on. In general, any consumed item that has sugar content contributes to that inventory. The large number of items an individual consumes suggests that this extension may be difficult.

Within the multiattribute framework, an attempt should be made to consolidate the predictive progress made by considering situational factors with the progress made by considering consumption histories. By capturing these two factors simultaneously and detailing the ways in which they might affect one another, we should move yet one step further toward explaining individual choice behavior.

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APPENDIX

The Introduction of Bias in Parameter Estimates by Ignoring Inventories

Consider the contribution of attribute j from item k at time T when inventory is accounted for and when it is not:

Inventory Accounted for:

$$P_{Tkj} = w_j [(I_{Tj} + X_{kj}) - \hat{X}_j]^2 (-1) \quad (2)$$

(See the original statement of Equation 2 for the explanation of the terms.)

Inventory not Accounted for:

$$\tilde{P}_{kj} = \tilde{w}_j [\tilde{X}_{kj} - \tilde{X}_j]^2 (-1) \quad (4)$$

where:

$\tilde{P}_{kj}$ = contribution of attribute j to the satisfaction that would result from consuming item k , not accounting for inventory

$\tilde{w}_j$ = importance weight for attribute j not accounting for inventory

$\tilde{X}_j$ = ideal level for attribute j not accounting for inventory

With the proposed estimation procedure, these expressions would be incorporated in the constraints of LINMAP once for each paired comparison in which item k is involved. Consider a specific paired comparison: item k preferred to

item k' . The related constraint accounting for inventory would imply that:

Accounting for Inventory:

$$\sum_{j=1}^J P_{Tkj} - \sum_{j=1}^J P_{Tk'j} + s_{kk'} \geq 0 \quad (5)$$

where

J = the number of relevant attributes

P_{Tkj} = contribution of attribute j to the satisfaction that would result from consuming item k at time T

$\sum_{j=1}^J P_{Tkj}$ = satisfaction that would result from consuming item k at time T

$s_{kk'}$ = artificial variable associated with the paired comparison (k, k') . (The sum of these artificial variables across all paired comparisons is the objective that LINMAP minimizes. It will equal zero if the paired comparison is satisfied. It will capture the magnitude of the violation otherwise.)

The constraint would take the following form if inventory were not accounted for:

Inventory not Accounted for:

$$\sum_{j=1}^J \tilde{P}_{Tkj} - \sum_{j=1}^J \tilde{P}_{Tk'j} + \tilde{s}_{kk'} \geq 0 \quad (6)$$

where tilded variables are the "not accounting for inventory" counterparts of variables just described.

To state the "accounting for inventory" constraint in terms of the parameters to be estimated, substitute Equation 2 into Inequality 5, yielding:

Accounting for Inventory:

$$\sum_{j=1}^J [w_j (X_{k'j}^2 - X_{kj}^2) - \underbrace{2w_j (\hat{X}_j - I_{Tj}) (X_{k'j} - X_{kj})}_{v_j}] + s_{kk'} \geq 0 \quad (7)$$

A similar substitution of Equation 4 into Inequality 6 and algebraic simplification yields the following constraint when inventory is not accounted for:

Inventory not Accounted for:

$$\sum_{j=1}^J [\tilde{w}_j (X_{k'j}^2 - X_{kj}^2) - \underbrace{2\tilde{w}_j \tilde{X}_j (X_{k'j} - X_{kj})}_{\tilde{v}_j}] + \tilde{s}_{kk'} \geq 0 \quad (8)$$

In the model not accounting for inventory, $\tilde{w}_j$ is estimated as the coefficient of $(X_{k'j}^2 - X_{kj}^2)$ and an artificial parameter,

$\tilde{v}_j = -2\tilde{w}_j \tilde{X}_j$, is estimated as the coefficient of $(X_{k'j} - X_{kj})$. The ideal point $\tilde{X}_j$ is then inferred to equal $-1/2 (\tilde{v}_j / \tilde{w}_j)$.

If the true process involves the evaluation of stimuli relative to an inventory, then reality is reflected in Inequality 7. Estimating the parameters with a model that does not account for inventory (Inequality 8), will yield $\tilde{w}_j$ and $\tilde{v}_j$ as coefficients of $(X_{k'j}^2 - X_{kj}^2)$ and $(X_{k'j} - X_{kj})$ respectively. From Inequality 7, then:

$$\tilde{w}_j = w_j \quad (9)$$

$$\tilde{v}_j = -2w_j(\hat{X}_j - I_{Tj}) \quad (10)$$

$$\tilde{X}_j = -1/2(\frac{\tilde{v}_j}{\tilde{w}_j}) = \hat{X}_j - I_{Tj} \quad (11)$$

From Equation 9, it is apparent that no bias is introduced into the importance weights by a model that does not account for inventory. Equation 11 shows that the ideal point estimated by a procedure not accounting for inventory is biased downward from the true ideal by an amount equal to the inventory at that point in time.

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